

Machine Learning

Week 4 Lecture

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Contents

- Intro to Support Vector Machines
- The objective function
- Kernel trick

Problem with Hard Margin Classifier

- Primary problem
 - complete lack of flexibility
- Specific failures of the hard margin approach
 - Failure on Non-Linearly Separable Data
 - Extreme Sensitivity to Outliers
 - Incompatibility with Large Datasets
 - Risk of Overfitting:
 - it tries to fit every training point perfectly, it often creates a "tight" boundary
 - Extreme Sensitivity to Outliers

The Complete Objective Function

- Optimization Problem

$$\min ||w||^2 \text{ s.t. } y_i(w^T x_i + b) \geq 1$$

- This is a hard maximum margin SVM classifier that assumes that the data is linearly separable.
- Not true in real world

- Soft Margin Classifier

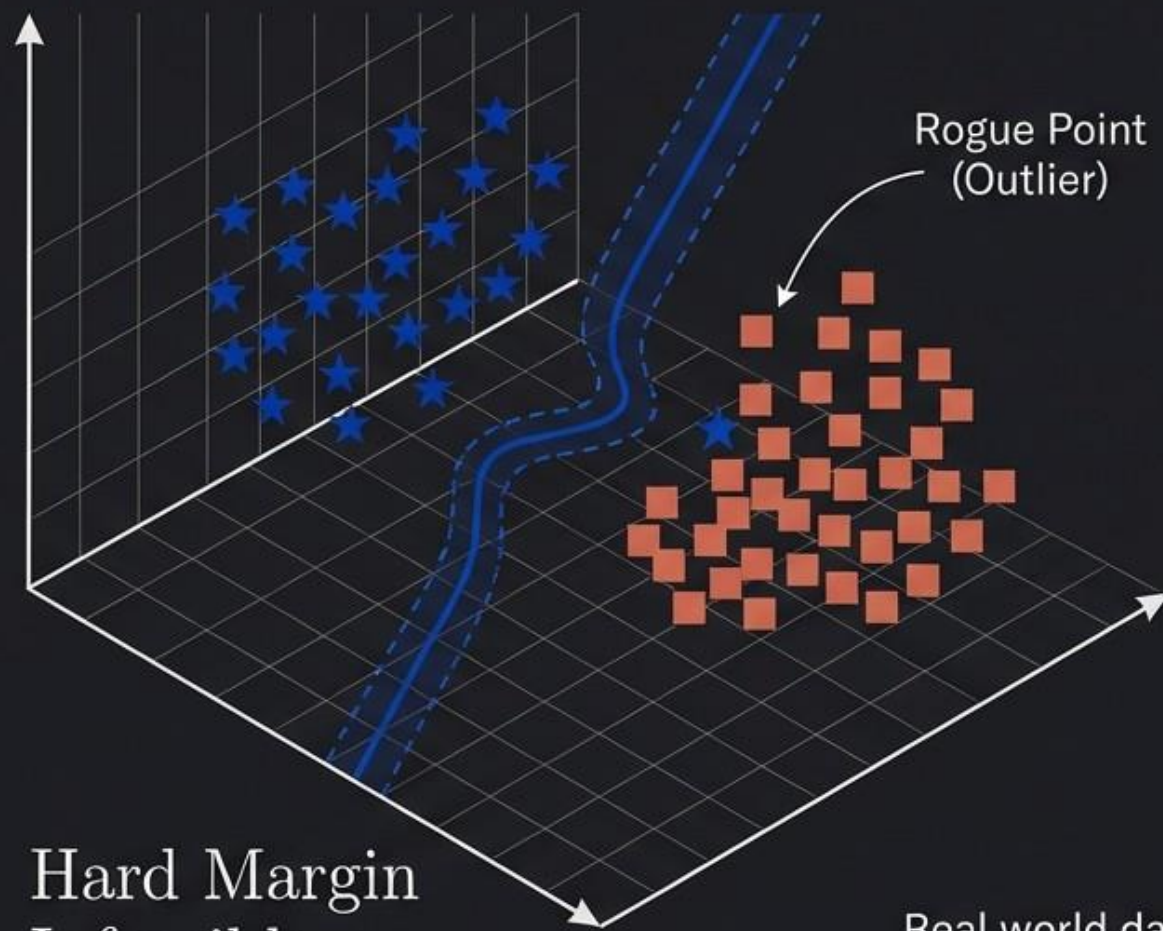
$$\min_{w,b,\xi} \frac{1}{2} ||w||^2 + C \sum_{i=1}^N \xi_i$$

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0$$

- C Penalize violations $C \sum_{i=1}^N \xi_i$

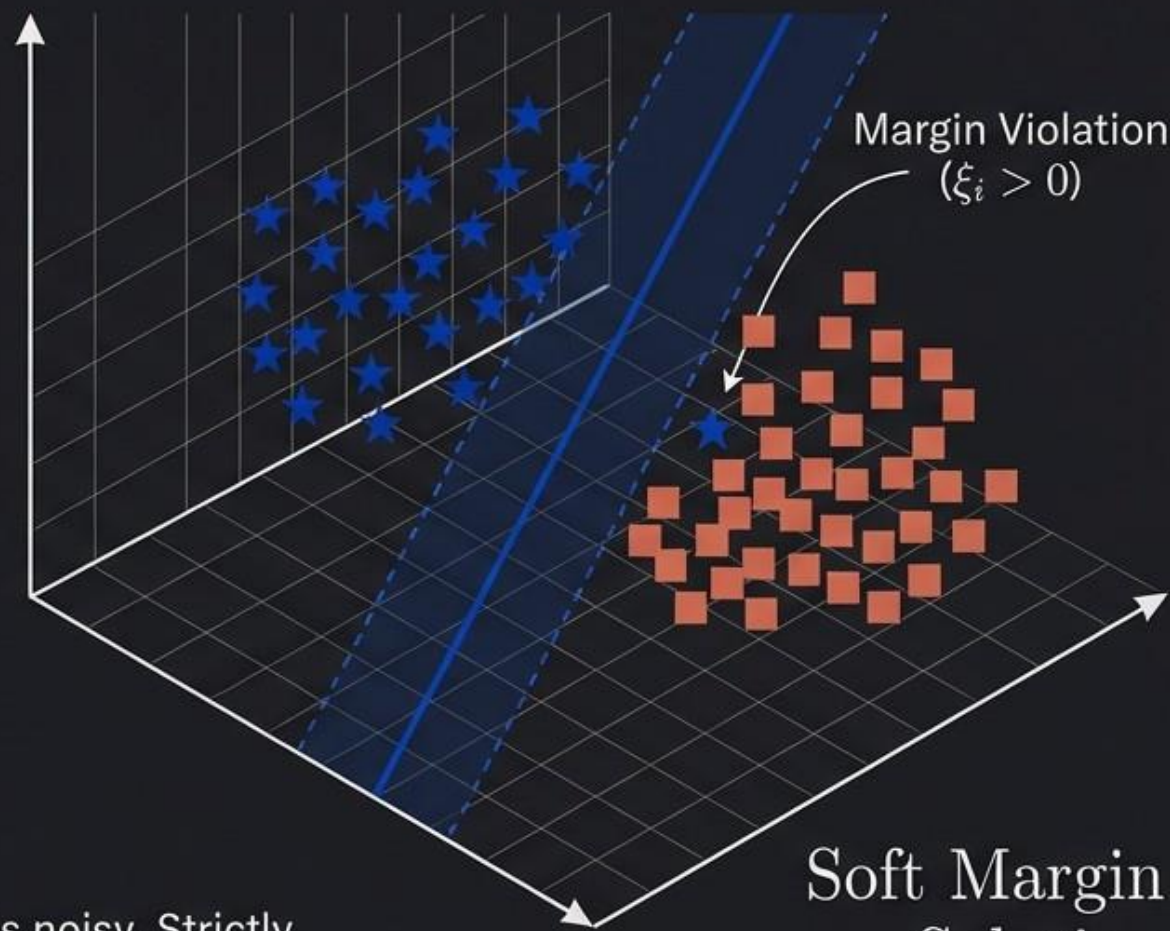
When Linearity Fails: The Soft Margin



**Hard Margin
Infeasible**

Strict separation fails due to noise.

Real world data is noisy. Strictly
enforcing separation breaks the math.
We must allow for calculated error.



**Soft Margin
Solution**

Allows calculated error for wider street.

Slack Variable

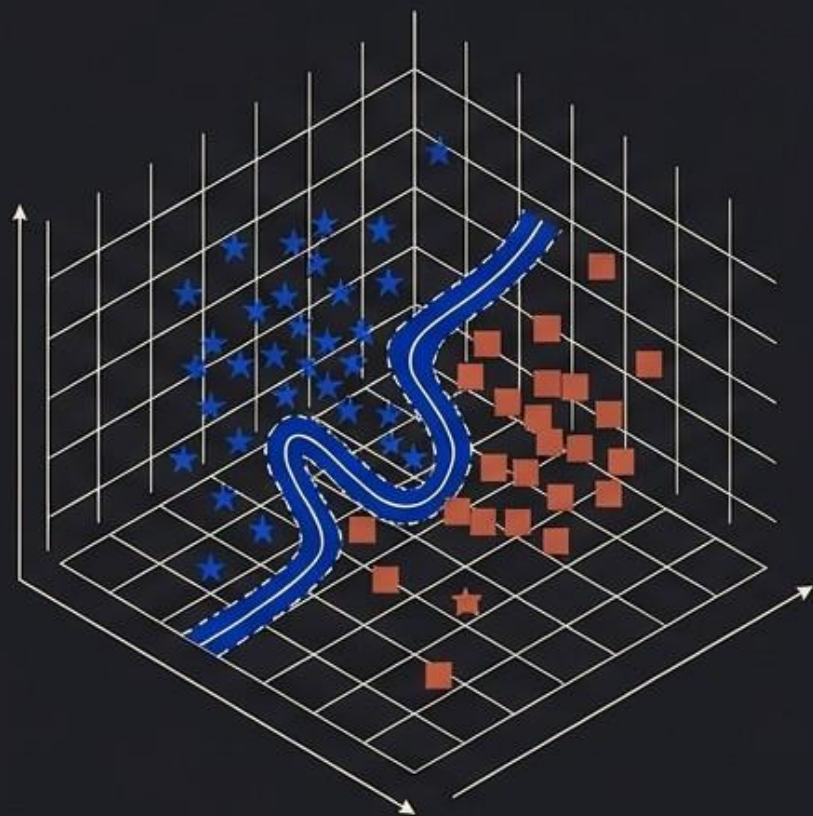
- Hinge Loss:

$$\xi_i = \max(0, 1 - y_i(w^T x_i + b))$$

- ξ_i : These represent the "degree of violation" for each point:
 - $\xi_i = 0$: Point is correctly classified and outside the margin.
 - $0 < \xi_i < 1$: Point is correctly classified but falls within the "street" (the margin).
 - $\xi_i > 1$: Point is misclassified (it has crossed over the decision boundary).
- **Geometrical Calculation:** During the training process, the optimization algorithm calculates the value of each ξ_i based on where the data point x_i actually lands relative to the margin:
 - $\xi_i = 0$: If the point is on the correct side of the margin.
 - $\xi_i = 1 - y_i(w^T x_i + b)$ If the point violates the margin

The Cost of Compromise: Hyperparameter C

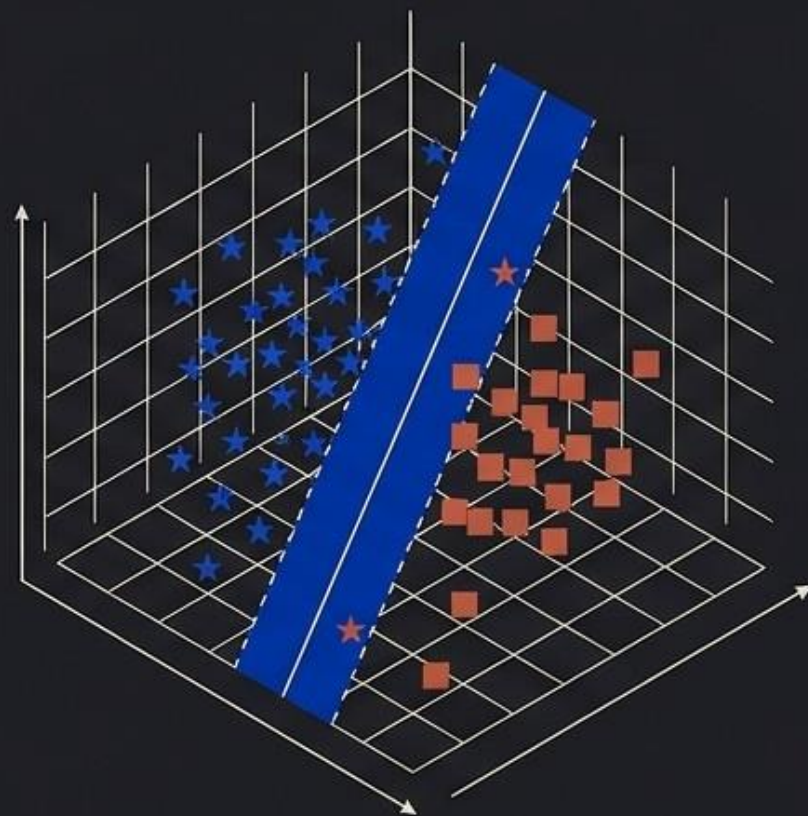
Large C (Strict)



Low Bias, High Variance
(Overfitting).

C is the penalty price we pay for misclassification.

Small C (Lenient)



High Bias, Low Variance
(Generalization).

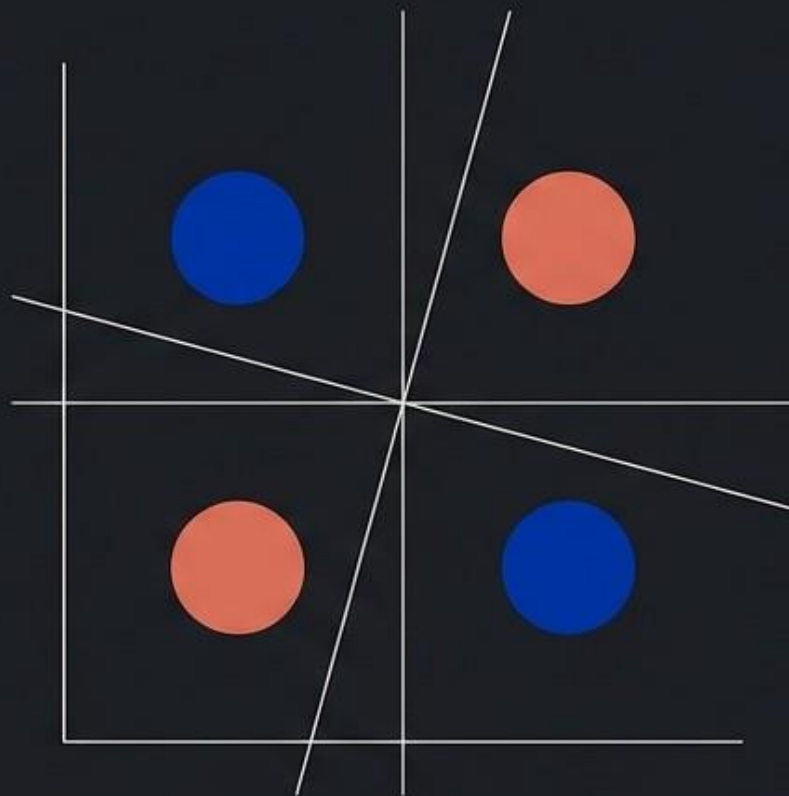
The Kernel Trick

- Problem: Non-linearly Separable Data
 - Example: XOR
 - Cannot separate with line in 2D.

- Trick: Map to Higher Dimension
 - Define feature map:

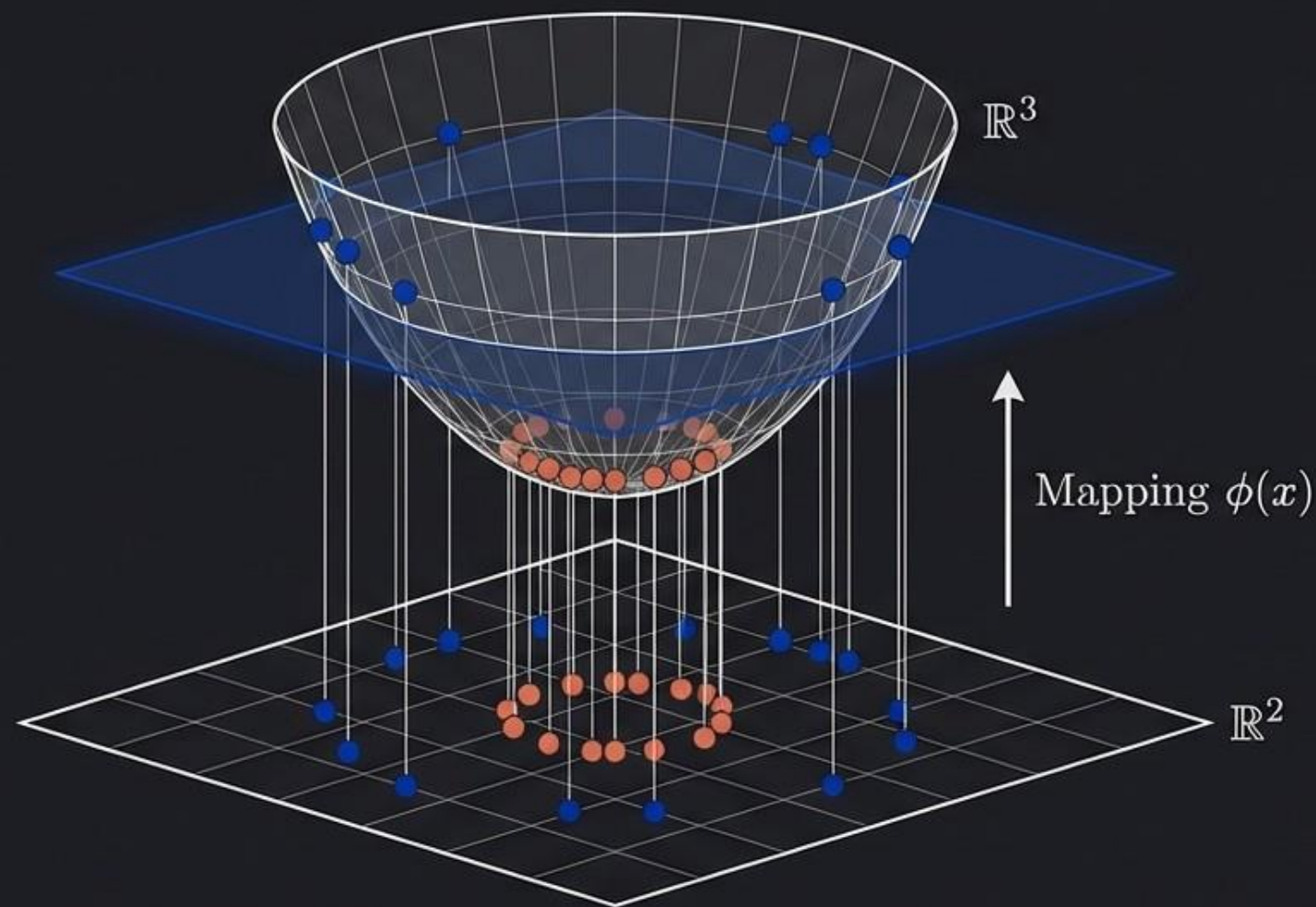
$$\phi(x) = (x_1, x_2, x_1^2 + x_2^2)$$

The Limits of Linearity: The XOR Problem



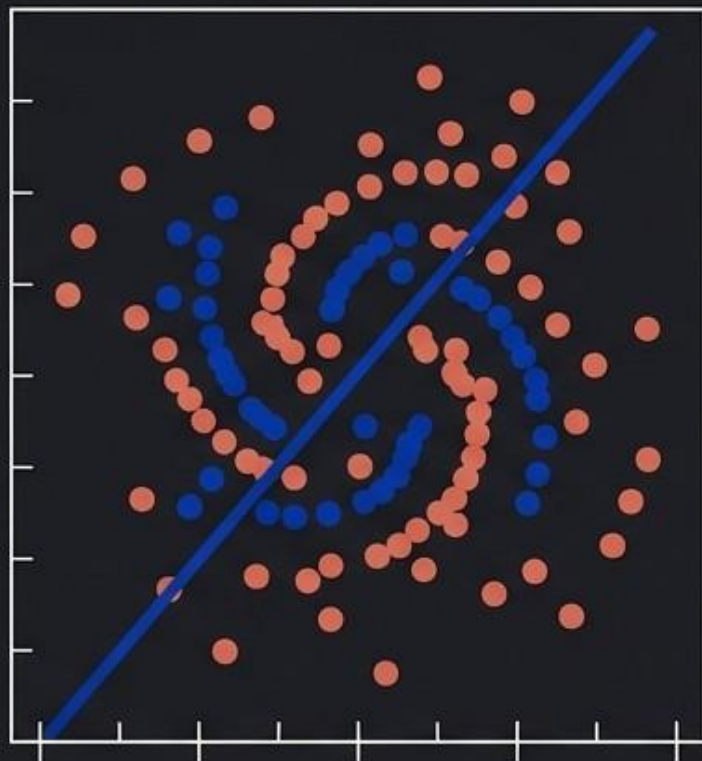
No single straight line can separate these classes.
In this dimension, the problem is unsolvable.

Escaping Flatland: The Feature Space

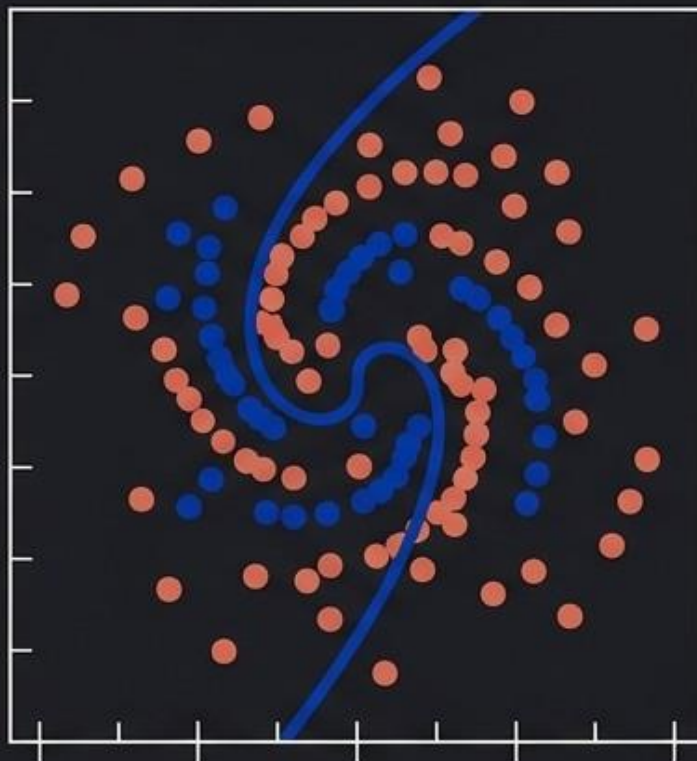


The Kernel Arsenal

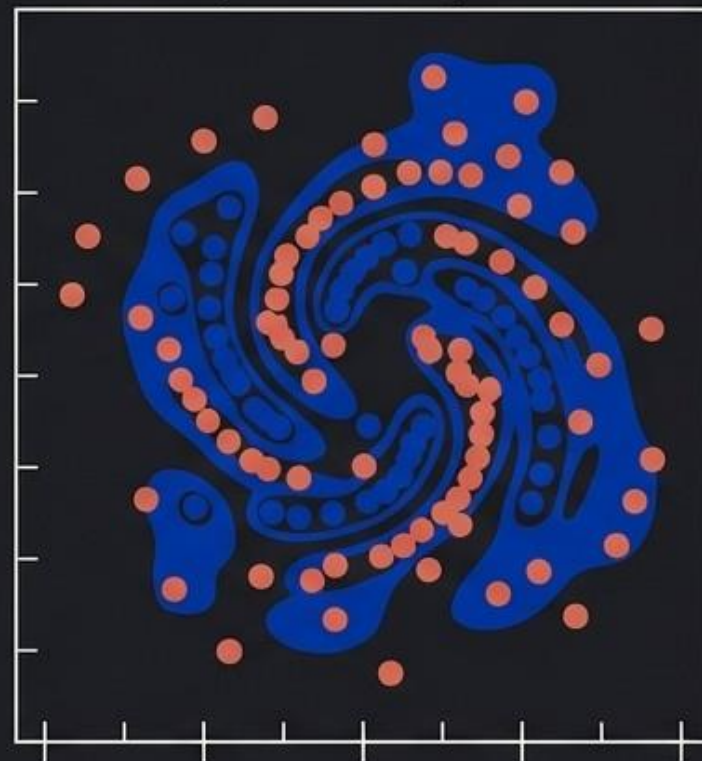
Linear Kernel



Polynomial Kernel



RBF (Gaussian) Kernel



SVM Dual Form

- Primal form: $\min \|w\|^2 \text{ s.t. } y_i(w^T x_i + b) \geq 1$

- We form Lagrangian:

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_i \alpha_i [y_i(w^T x_i + b) - 1]$$

- Take derivatives and eliminate w and b

- We get: $w = \sum_i \alpha_i y_i x_i$

- Substitute back:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$\sum_i \alpha_i y_i = 0$$

$$\alpha_i \geq 0$$

- This is the dual form of SVM

- Important Note:

– Data appears only as **dot products** $x_i^T x_j$

SVM with Kernel

- Kernel Trick
 - Map data to a higher dimension $\phi(x)$
 - Compute and replace dot product $\phi(x_i)^T \phi(x_j)$
 - Where:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

- The dual becomes:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

Math Kernel Example

$$\phi(x) = (x, x^2)$$

$$x_i = 2, \quad x_j = 3$$

$$\phi(2) = (2, 4)$$

$$\phi(3) = (3, 9)$$

$$\phi(2)^T \phi(3) = 2 \cdot 3 + 4 \cdot 9 = 6 + 36 = 42$$

Common Kernels

Linear Kernel

$$K(x, z) = x^T z$$

Polynomial Kernel

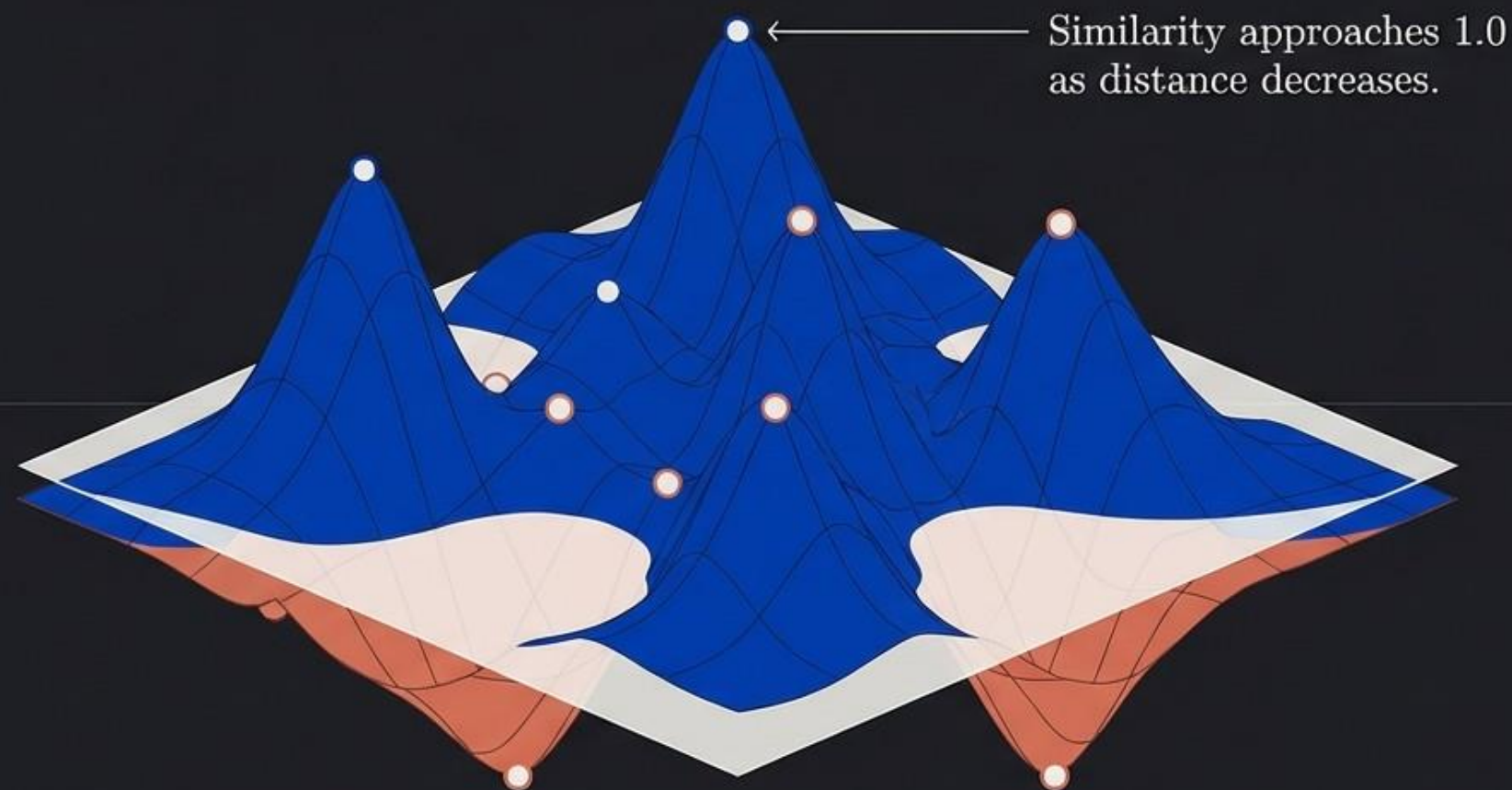
$$K(x, z) = (x^T z + 1)^d$$

RBF Kernel

$$K(x, z) = \exp(-\gamma \|x - z\|^2)$$

RBF = infinite-dimensional mapping.

Infinite Dimensions: The RBF Kernel



The Radial Basis Function lifts data into infinite dimensions, wrapping a surface around the points like shrink-wrap.